

Resilience in Distributed Observer based on Trust for Connected Vehicles

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Abstract: This paper proposes a trust-aware distributed observer for vehicle platoons that maintains resilient state estimation under cyberattacks. A behavioral divergence metric evaluates the reliability of shared data, forming a dynamic neighbor set used to adapt observer weight gains. Stability conditions are derived via Lyapunov analysis. Simulations under bogus, replay, and DoS attacks demonstrate robust performance and stable platoon behavior.

Keywords: Distributed observers, secure estimation, lyapunov stability, cyber-attack.

1. INTRODUCTION

The rapid development of autonomous and connected vehicles has introduced new opportunities to improve road safety, traffic flow, and fuel efficiency. In vehicle platoons—where multiple vehicles coordinate through vehicle-to-vehicle (V2V) communication—accurate distributed state estimation is essential for maintaining stability and efficiency. Each vehicle must reconstruct both its own and others' states using locally sensed data and information exchanged with neighbors. However, this distributed structure also exposes the system to cyber and communication attacks that can disrupt coordination or compromise state estimation.

Traditional estimation and control methods generally assume all vehicles are cooperative and trustworthy, which is unrealistic in adversarial environments. Malicious or faulty agents can transmit falsified information, causing estimation errors that propagate through the network. Consequently, resilient estimation strategies capable of identifying and isolating untrustworthy data have become a key research direction in cyber-physical systems (CPS) and distributed multi-agent systems (DMAS).

Existing work on secure estimation and fault detection can be categorized into three main families:

- Observer-based detection: compares model-predicted states with sensor measurements to identify anomalies. These methods are effective for detecting deviations but may propagate errors if compromised neighbors provide false data.
- Consensus-based detection: relies on cross-validation among agents to identify inconsistent information. While robust under the assumption that most agents

are trustworthy, their performance degrades under large-scale coordinated attacks.

- Trust-based detection: assigns reliability scores to neighboring agents based on behavioral or statistical consistency. Physical-signal approaches use device fingerprints for authentication, while statistical approaches compare communicated data against locally observed behaviors. These methods improve resilience by filtering unreliable inputs before consensus or estimation steps, but they typically require significant computational resources or long-term historical data-making real-time adaptation challenging for dynamic platoons.

Mitigation strategies in the literature often depend on attack detection results. Some switch to fallback modes (e.g., CACC to ACC), while others incorporate trust scores directly into estimation and control to down-weight suspicious data. The latter provides finer control adaptation but relies heavily on accurate and timely trust evaluation.

Motivation and Contributions: To overcome these limitations, we propose a trust-aware distributed observer framework for vehicle platoons. The framework continuously evaluates the trustworthiness of communicated data by comparing reported and predicted behaviors, dynamically adjusting observer weights according to each neighbor's trust score. This mechanism ensures reliable estimation even under cyber-attacks or communication faults.

The main contributions are:

- A formal definition of the resilient distributed state estimation problem for connected vehicles under adversarial conditions.
- A behavioral trust model that quantifies each agent's reliability and constructs a dynamic trusted neighbor set.
- A trust-informed distributed observer with provable stability guarantees, validated through simulations under multiple attack scenarios.

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By integrating detection, mitigation, and estimation within a unified trust-based layer, the proposed method improves the resilience and stability of cooperative vehicle systems. The remainder of this paper is organized as follows. Section 2 describes the vehicle model. Section 3 introduces the distributed observer design. Section 4 presents the trust framework and trust-weight-based adaptation. Section 5 presents the weight design and the stability analysis. Section 6 validates the proposed approach through simulation studies under multiple attack scenarios, followed by conclusions and future work in Section 7

1.1 Notation

For a square matrix $M \in \mathbb{R}^{n \times n}$, $\rho(M)$ denotes the spectral radius of M . $e_q(k) = \begin{bmatrix} 0, \dots, 1, \dots, 0 \end{bmatrix}$ is a vector of the canonical basis of $\mathbb{R}^q$.

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an undirected graph with N nodes, where:

- $\mathcal{V} = \{1, 2, \dots, N\}$ is the set of nodes.
- $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges.
- The weighted adjacency matrix $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{N \times N}$ of the graph $\mathcal{G}$ is an $N \times N$ matrix where:

$$[a_{ij}] = \begin{cases} a_{ij}, & \text{if } (i, j) \in \mathcal{E}, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

- The Laplacian matrix of graph $\mathcal{G}$ is denoted as $\mathcal{L} = [l_{ij}] \in \mathbb{R}^{N \times N}$ where

$$l_{ij} = \begin{cases} \sum_{k=1}^N a_{ik}, & \text{if } i = j, \\ -a_{ij}, & \text{otherwise.} \end{cases} \quad (2)$$

- The set of the neighbors of the node i is defined by $\mathcal{N}_i = \{j \in \mathcal{V} \mid (j, i) \in \mathcal{E}\}$.

2. VEHICLE MODEL

2.1 Dynamic Model of Single Vehicle

We consider the discrete-time longitudinal dynamics of each vehicle $i \in \mathcal{V}$:

$$x_i(t+1) = A_i x_i(t) + B_i u_i(t), \quad (3)$$

where $x = [s_i \ v_i \ a_i]^\top$ and u_i denotes position, velocity, acceleration, and control input of vehicle i , respectively; The system matrices are given by

$$A_i = \begin{bmatrix} 1 & T_s & \frac{T_s^2}{2} \\ 0 & 1 & T_s \\ 0 & 0 & 1 - \frac{T_s}{\tau_i} \end{bmatrix}, B_i = \begin{bmatrix} 0 \\ 0 \\ \frac{T_s}{\tau_i} \end{bmatrix}, C_i = I_3$$

where T_s is the sampling time and the constant $\tau_i > 0$ characterizes the engine's time lag. Since all vehicles in the platoon are usually of similar type, the variation among A_i and B_i is small. Therefore, we introduce a nominal dynamic model (A_b, B_b) representing a baseline vehicle, and treat the modeling differences as an additive uncertainty term $\Delta_i(t)$.

$$x_i(t+1) = A_b x_i(t) + B_b u_i(t) + \Delta_i(t), \quad (4)$$

where $\Delta_i(t)$ captures the modeling differences and uncertainties.

2.2 Dynamic Model of Platoon

Define the collective state as $x = \text{col}(x_{(1)}, \dots, x_{(N)})$, the state-space equation of the platoon is

$$\begin{cases} x(t+1) = (I_N \otimes A_b)x(t) + (I_N \otimes B_b)u(t) + \Delta(t), \\ y_i(t+1) = Cx(t), \end{cases} \quad (5)$$

where $u = \text{col}(u_1, \dots, u_N)$ is the collective control input $\Delta = \text{col}(\Delta_1, \dots, \Delta_N)$ group unmodeled dynamics and external disturbances and $C = \begin{bmatrix} 0, \dots, \underbrace{C_i}_{i\text{th}}, \dots, 0 \end{bmatrix}$.

Assumption 1. We assume that $\Delta(t)$ is bounded, i.e., $\|\Delta(t)\| \leq \bar{\Delta}$.

Since each vehicle measures its own full state through onboard sensors, ensuring that the pair (A_i, C_i) is observable. However, in the context of the platoon dynamic (5), the pair $((I_N \otimes A_b), C)$ remain unobservable, motivating the need for distributed estimation through inter-vehicle communication.

3. DISTRIBUTED OBSERVER

The distributed observer serves as an effective means to estimate the entire state x of the platoon. In this section, we will briefly revisit existing findings on the distributed observer for the platoon. Following this, the main concept of this paper will be introduced.

To achieve estimation of all vehicles' states, two estimation layers are employed: a Local Observer ($\mathcal{LO}$) that uses onboard sensors, and a Distributed Observer ($\mathcal{DO}$) that fuses information exchanged through communication links.

This distributed state estimation equation can be summarized as shown in (8), where

- $\hat{x}_0^{(i)}(t)$ is the state of the local observer $\mathcal{LO}_i$, which will try to track the states of the i th vehicle. It means that

$$\lim_{t \rightarrow \infty} (\hat{x}_0^{(i)}(t) - x^{(i)}(t)) = 0. \quad (6)$$

- F_i is the gain matrix of the local observer.
- $\mathcal{N}_i$ is the neighbors of the i th vehicle according to the communication graph $\mathcal{G}$, which will be explained in Remark 3.
- $w_{il}^{(j)}$, $l \in \mathcal{N}_i^{(j)}$ is the gain scale to be designed based on the communication topology.
- $\hat{x}_i^{(j)}(t)$ is the state of the distributed observer $\mathcal{DO}_i^{(j)}$, which will try to track the states of the platoon. It means that

$$\lim_{t \rightarrow \infty} (\hat{x}_i(t) - x(t)) = 0, \quad (7)$$

where $\hat{x}_i(t) = \text{col}(\hat{x}_i^{(1)}(t), \dots, \hat{x}_i^{(N)}(t))$ is the collective states of all distributed observer.

- $w_{i0}^{(j)}$ is the gain scale to be designed, which is such that

Remark 1. (Explanation of observer structure 8). The distributed observer $\mathcal{DO}_i^{(j)}$ is designed to estimate the state

$$\mathcal{LO}_i : \hat{x}_0^{(i)}(t+1) = A_b \hat{x}_0^{(i)}(t) + B_b u_i(t) + F_i \left(y_i(t) - C_b \hat{x}_0^{(i)}(t) \right), \quad (8a)$$

$$\mathcal{DO}_i^{(j)} : \hat{x}_i^{(j)}(t+1) = A_b \left(\hat{x}_i^{(j)}(t) + \sum_{l \in \mathcal{N}_i} w_{il}^{(j)}(t) \left(\hat{x}_l^{(j)}(t) - \hat{x}_i^{(j)}(t) \right) + w_{i0}^{(j)}(t) \left(\hat{x}_0^{(j)}(t) - \hat{x}_i^{(j)}(t) \right) \right) + B_b \hat{u}_j(t), \quad (8b)$$

of every vehicle $j \in \mathcal{V}$ in the platoon from the perspective of vehicle i . The term $\sum_{l \in \mathcal{N}_i} w_{il}^{(j)} \left(\hat{x}_l^{(j)}(t) - \hat{x}_i^{(j)}(t) \right)$ leverages communication with neighbors $l \in \mathcal{N}_i$ to adjust $\hat{x}_i^{(j)}(t)$ towards a consensus among neighboring estimates of vehicle j 's state. The weights $w_{il}^{(j)}$ determine the influence of each neighbor's estimate, promoting consistency across the platoon, which will be designed later.

The consensus term alone has a key limitation, it cause all vehicles' estimates of a given state $\hat{x}_i^{(j)}(t)$ to converge to a common value. This is theoretically sound only for an ideal platoon with perfectly aligned states. However, practical disturbances, initial condition errors, and imperfect models break this uniformity, rendering a pure consensus estimate inaccurate for any individual vehicle's true state.

To mitigate this, the local observer $\mathcal{LO}_i$ provides an accurate estimate of vehicle i 's own state, $\hat{x}_0^{(i)}(t)$. The additional anchoring term $w_{i0}^{(j)} \left(\hat{x}_0^{(j)}(t) - \hat{x}_i^{(j)}(t) \right)$ injects this trusted reference, allowing each vehicle to correct model errors and maintain accurate peer estimates.

Remark 2. If a host forms $\hat{u}_j(t)$ (e.g., from a known controller law or an input estimator), the mismatch

$$\delta u_j(t) = u_j(t) - \hat{u}_j(t) \quad (9)$$

will be treated as part of the disturbance via Δ . We do not require $\hat{u}_j(t)$ for the DO to run; using it only tightens bounds.

Remark 3. (Explanation of the communication graph). Since

each distributed observer $\mathcal{DO}_i^{(j)}$ requires reference information from the local observer $\mathcal{LO}_j$. To formally include this reference, we define a virtual node labeled as node 0 that connects to the vehicle j , where $j \in \mathcal{V}$ and its neighbors $\mathcal{N}_j$ according to the original graph $\mathcal{G}$. The resulting virtual communication graph $\bar{\mathcal{G}}^{(j)} = \{\bar{\mathcal{V}}, \bar{\mathcal{E}}^{(j)}\}$ is constructed as:

$$\begin{aligned} \bar{\mathcal{V}} &= \mathcal{V} \cup \{0\}, \\ \bar{\mathcal{E}}^{(j)} &= \mathcal{E} \cup \{(0, j) | j \in \mathcal{V}\} \cup \{(0, k) | k \in \mathcal{N}_j\}. \end{aligned} \quad (10)$$

The adjacency matrix $\bar{\mathcal{A}}^{(j)} \in \mathbb{R}^{(N+1) \times (N+1)}$ corresponding to $\bar{\mathcal{G}}^{(j)}$ is defined as:

$$\bar{\mathcal{A}}^{(j)} = \begin{bmatrix} 0 & 0 \\ w_0^{(j)} & \mathcal{A} \end{bmatrix}, \quad (11)$$

where the vector $w_0^{(j)} = \text{col} \left(w_{10}^{(j)}, \dots, w_{N0}^{(j)} \right) \in \mathbb{R}^N$ defines the connection strengths from node 0 to the distributed observers.

This virtual communication graph provides a unified representation for integrating local estimation (through node 0) and neighbor-based consensus (through graph $\mathcal{G}$), ensuring that every vehicle's local observer acts as a trusted anchor for distributed state estimation.

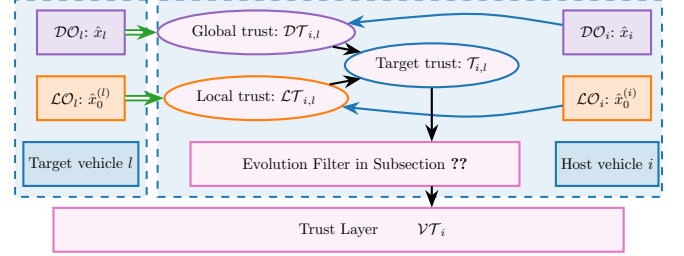


Fig. 1. Trust framework.

3.1 Main Objectives

Traditional distributed observers use fixed weights w_{il} that assume all data sources are trustworthy. However, in a connected vehicle platoon, compromised nodes may send falsified information. To address this, the proposed design incorporates trust-based weighting, allowing each vehicle to dynamically adjust its observer weights according to the reliability of received information.

4. TRUST SCORE FRAMEWORK

To ensure resilient estimation, each vehicle continuously evaluates the reliability of the data it receives. This is achieved through a multi-layer trust framework that quantifies how much a host vehicle i trusts another vehicle l . The overall structure of this trust framework is illustrated in Fig. 1. In this context, we define the following types of trust:

- **Local Trust** $\mathcal{LT}_{i,l}$: evaluates the reliability of the local observer $\mathcal{LO}_l$ data from vehicle l based on consistency and timeliness.
- **Distributed Trust** $\mathcal{DT}_{i,l}$: measures how well the global estimates of l agree with those of i .
- **Target Trust** $\mathcal{T}_{i,l}$: Combining the Local Trust and Global Trust, the host vehicle i can evaluate the trust score for each target vehicle at time t .
- **Vehicle Trust** $\mathcal{VT}_i$: Smooth evolution of Target Trust scores over time.

In the following subsections, we will explain how to calculate the trust. After that, we will provide more details on how to determine vehicle trust ($\mathcal{VT}_i$) and how to utilize it effectively.

4.1 Data Validity Indicator

Before trust computation, every received V2V packet data is validated for authenticity and freshness:

$$\delta_{i,l}(t) = \begin{cases} 1, & \text{if message is authenticated and fresh,} \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

Only packets satisfying $\delta_{i,l}(t) = 1$ use in the following trust evaluation process.

4.2 Local Trust

The Local Trust $\mathcal{LT}_{i,l}$ quantifies the confidence that the host vehicle i places in the local observer output of a neighboring vehicle $l \in \mathcal{N}_i$. It is based on three physical-consistency indicators computed from measurable states: velocity, distance, and acceleration. When data are missing, a hold-and-decay rule maintains continuity.

Velocity consistency The host estimates the expected velocity of the platoon leader using its last valid packet as a reference value:

$$v_{\text{ref}}(t) = v_L(t - t_{\text{elap}}) + t_{\text{elap}} \cdot a_L(t - t_{\text{elap}}), \quad (13)$$

Then, the mismatch score of the target vehicle i with the reference leader can be calculated as

$$v_{i,L} = \begin{cases} \max \left(1 - \left| \frac{\hat{v}_0^{(l)} - v_{\text{ref}}}{v_{\text{ref}}} \right|, 0 \right), & \text{if } v_{\text{ref}} > 0, \\ \max \left(1 - |\hat{v}_0^{(l)}|, 0 \right), & \text{otherwise,} \end{cases} \quad (14)$$

Similarly, the host vehicle $\hat{v}_0^{(i)}$ compares with its neighbors $\hat{v}_0^{(l)}$, the following mismatch score is:

$$v_{i,H} = \begin{cases} \max \left(1 - \left| \frac{\hat{v}_0^{(l)} - \hat{v}_0^{(i)}}{\hat{v}_0^{(i)}} \right|, 0 \right), & \text{if } \hat{v}_0^{(i)} > 0, \\ \max \left(1 - |\hat{v}_0^{(l)}|, 0 \right), & \text{otherwise,} \end{cases} \quad (15)$$

The velocity mismatch is computed as a weighted average of two estimates:

$$\iota_{i,l}^{\text{velocity}} = ((1 - \sigma)v_{i,L} + \sigma v_{i,H}) \quad (16)$$

where the weighted σ is selected based on the index position of the target vehicle l . If the target is following the host vehicle i , then $\sigma \in (0, 0.3)$. If the target is leading, $\sigma \in (0.7, 1)$.

Distance consistency The host vehicle i calculates the distance with respect to the target l as :

$$d_{i,l} = \hat{s}_{x,0}^{(l)} - \hat{s}_{x,0}^{(i)} - L,$$

where L is the length of the vehicle. So the mismatch between the measured distance d from its sensors and the calculated data $d_{k,i}$ is

$$\iota_{i,l}^{\text{distance}} = \max \left(1 - \left| \frac{d_{i,l} - d}{d} \right|, 0 \right). \quad (17)$$

Assumption 2. When the vehicle is connected, it knows basic information about its neighbors, such as the type and length of the vehicle.

Acceleration consistency The host vehicle can calculate the expected relative acceleration as

$$a_{\text{recv}}^{\text{rel}} = \hat{a}_0^{(l)} - \hat{a}_0^{(i)}, \quad a_{\text{expect}}^{\text{rel}} = \frac{v_{\text{rel}}(t) - v_{\text{rel}}(t - n)}{n},$$

where $v_{\text{rel}} = \hat{v}_0^{(l)} - \hat{v}_0^{(i)}$ and n defines the averaging window used to smooth noise. To account for physical context,

the mismatch is normalized by both distance and relative velocity:

$$\iota_{i,l}^{\text{accel}} = \max \left(1 - \frac{|a_{\text{recv}}^{\text{rel}} - a_{\text{expect}}^{\text{rel}}|}{a_{\text{th}} \left(1 + \frac{d_{i,l}}{d_{\text{norm}}} \right) \left(1 + \frac{|v_{\text{rel}}|}{v_{\text{norm}}} \right)}, 0 \right) \quad (18)$$

where

- a_{th} is a nominal acceleration tolerance ($\approx 0.5 - 1m/s^2$),
- $d_{i,l}$ is the inter-vehicle distance,
- d_{norm} and v_{norm} are normalization constants reflecting typical spacing and speed in the platoon.

This formulation tightens the trust threshold when vehicles are close or moving fast where inconsistencies are more critical and relaxes it when they are far apart or nearly stationary. It yields smoother, dimensionally consistent trust updates and reduces false trust drops caused by small sensor fluctuations.

Fusion and update rule We combine all three indicators into an integrity measure $\mathcal{LT}_{i,l}$ indicates the trust of local data of the host vehicle i in the target vehicle l at the time t :

$$\mathcal{LT}_{i,l}(t) = \begin{cases} f(\iota_{i,l}^{\text{velocity}}, \iota_{i,l}^{\text{distance}}, \iota_{i,l}^{\text{accel}}) & \text{if } \delta_{i,l}(t) = 1 \\ (1 - \lambda_h) \mathcal{LT}_{i,l}(t - 1), & \text{otherwise} \end{cases} \quad (19)$$

where f is a fusion function that combines the three scores. The parameter λ_h defines the slow decay during temporary losses.

Remark 4. Function f can be a weighted average or a minimum function, depending on the desired sensitivity to individual indicators. The decay λ_h can be varied based on how many consecutive packets are missed. This prevents a single missed packet from instantly reducing trust to zero, while still penalizing prolonged disconnection.

4.3 Distributed Trust

Same concept of Local Trust but in the context of the distributed data. We denote $\mathcal{DT}_{i,l}$ represents how the host vehicle i trusts the distributed estimation received from target vehicle l , which is defined as

$$\mathcal{DT}_{i,l} = \begin{cases} \gamma_{i,l}^{\text{host}} \cdot \gamma_{i,l}^{\text{local}} & \text{if } \delta_{i,l}(t) = 1 \\ (1 - \lambda_h) \mathcal{DT}_{i,l}(t - 1), & \text{otherwise} \end{cases} \quad (20)$$

where γ_l^{host} and $\gamma_{i,l}^{\text{local}}$ are applied to check if the global estimates from neighbor l are consistent with the host's and the local measurements, respectively; Both of them will be illustrated in the following.

Consistency with host's global estimate To assess whether the global estimate from the target l is trustworthy, the host can compare it directly to its own global estimate. We use the Mahalanobis distance, which normalizes discrepancies based on covariance:

$$R_{i,l}^{(j)} = \left(\hat{x}_i^{(j)} - \hat{x}_l^{(j)} \right)^T \Sigma_{\text{host}}^{-1} \left(\hat{x}_i^{(j)} - \hat{x}_l^{(j)} \right), \quad (21)$$

where Σ_{local} is the diagonal covariance matrix which captures variance and correlations between each state channel defined as

$$\Sigma_{\text{local}} = \text{diag} (\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2), \quad (22)$$

where n is the number of states in the vehicle, and σ_k^2 is the variance of discrepancies about the different states.

And then, add up these differences to get the total discrepancy score across the platoon as the following

$$D_{i,l} = \sum_{j=1}^N d_{i,l}^{(j)}. \quad (23)$$

A small $D_{i,l}$ means the global estimate from vehicle l is close to the host's, suggesting higher trustworthiness.

Based on the total discrepancy score across the platoon, the trust score can be converted to be

$$\gamma_{i,l}^{\text{host}} = \exp(-D_{i,l}). \quad (24)$$

where $\gamma_{i,l}^{\text{host}} \in (0, 1]$. A value near 1 means high trust (small discrepancy), while a value near 0 means low trust (large discrepancy).

Consistency with host's local measurement Host vehicle i have relative measurement to nearby vehicles (e.g., distance and relative velocity to a predecessor and a follower) and use that to checks if the distributed estimates from target l are consistent with these measurements.

Let $y_{i,j}(t) \in R^{m_{ij}}$ is the relative measurement of vehicle i to vehicle j (e.g., $y_{i,j}(t) = s_j(t) - s_i(t)$ for relative position).

The distributed estimate of target l should satisfy expectively:

$$H(\hat{x}_l^{(j)}(t) - \hat{x}_l^{(i)}(t)) \approx y_{i,j}(t).$$

where $H \in R^{m_{ij} \times n}$ is the measurement matrix that extracts the relevant states (e.g., position) from the state vector.

Define the error between the global estimate and the local measurement as

$$\varepsilon_{i,l}^{(j)}(t) = (\hat{x}_l^{(j)}(t) - \hat{x}_l^{(i)}(t)) - y_{i,j}(t), \quad (25)$$

Similar with the (21), by applying the covariance matrix Σ_{local} , we can get

$$\varepsilon_{i,l}^{(j)}(t) = (\varepsilon_{i,l}^{(j)}(t))^{-1} \Sigma_{\text{local}}^{-1} \varepsilon_{i,l}^{(j)}(t). \quad (26)$$

Then, the total consistency error can be get by summing up the errors over all measurable vehicles:

$$E_{i,l}(t) = \sum_{j \in \mathcal{N}_i} E_{i,l}^{(j)}(t). \quad (27)$$

A higher $E_{i,l}(t)$ suggests that the distributed estimate of target l contradicts with local sensor relative measurements.

Convert the error into a trust factor as

$$\gamma_{i,l}^{\text{local}}(t) = \exp(-E_{i,l}(t)), \quad (28)$$

where $\gamma_{i,l}^{\text{local}}(t) \in (0, 1]$, with lower values indicating poor consistency.

Remark 5. We can estimated covariance Σ_{host} and Σ_{local} by collecting the differences in vehicle i from a window of recent received data from l in normal condition. This avoids the need for each node to transmit its own covariance.

4.4 Target Trust and Temporal Evolution

We can combine the global trust and the local trust to get one single trust of the target l by following :

$$\mathcal{T}_{i,l}(t) = \mathcal{L}\mathcal{T}_{i,l} \cdot \mathcal{D}\mathcal{T}_{i,l}. \quad (29)$$

$\mathcal{T}_{i,l}(t)$ indicates the quality of target l data at the time t , that will facilitate the decision-making process for the host vehicle.

Because this value can vary abruptly, a temporal-evolution mechanism is introduced to obtain a smoother, interpretable estimate.

Discretization into trust quality We define a five-level trust quality among $q = 5$ categories: *Unreliable*, *Poor*, *Acceptable*, *Good*, and *Excellent*. When the host vehicle i computes a trust value $\mathcal{T}_{i,l}(t) \in [0, 1]$ for a neighboring vehicle l , the result is discretized into a one-hot vector $r_{i,l}(t) \in \mathbb{R}^q$. For example, if $\mathcal{T}_{i,l}(t) = 0.72$, it falls into the "Good" category (4th level), so $r_{i,l}(t) = [0, 0, 0, 1, 0]^\top$.

Accumulated trust history Over time, the accumulated trust vector $R_{i,l}(t)$ for target vehicle l is updated using a weighted aggregation of past interactions:

$$R_{i,l}(t+1) = (1 - \mathcal{T}_{i,l}(t) \cdot \beta) \cdot R_{i,l}(t) + r_{i,l}(t), \quad (30)$$

where $\beta \in [0, 1]$ as a tunable weight controls how quickly past experiences decay.

Remark 6. The choice of β is crucial for effectively capturing the dynamics of trust over time. A higher β places more emphasis on recent observations, allowing the trust score to adapt quickly to changes in behavior. Conversely, a lower β gives more weight to historical data, resulting in a smoother trust evolution that is less sensitive to transient fluctuations.

Normalized trust distribution Finally, at the current step t (in the following, we omit t for simplifying), a normalized trust distribution vector $S_{i,l}(k)$ is computed by incorporating an a priori distribution α and a confidence parameter C :

$$S_{i,l}(k) = \frac{R_{i,l}(k) + C \cdot \alpha(k)}{C + \sum_{j=1}^q R_{i,l}(j)}, \quad (31)$$

which provides a probabilistic interpretation of the trustworthiness of the target vehicle l , blending historical observations with prior beliefs to produce a robust and adaptive trust estimate.

Expected trust level The target vehicle trust score $\mathcal{V}\mathcal{T}_{i,l}$ is then calculated as:

$$\mathcal{V}\mathcal{T}_{i,l} = \sum_{j=1}^q \frac{j-1}{q-1} \cdot S_{i,l}(j). \quad (32)$$

While $\mathcal{V}\mathcal{T}_{i,l}$ provides a stable, pairwise evaluation between directly connected vehicles, a more global view is required for the observer to weigh all available information. This motivates the definition of a generalized trust vector, introduced next.

4.5 Generalized Trust Vector

The target trust value $\mathcal{V}\mathcal{T}_{i,l}$ obtained in the previous subsection quantifies how much the host vehicle i trusts a

specific neighbor l . However, for the distributed observer to operate reliably, each vehicle must maintain an opinion about all other vehicles in the platoon, including those beyond its direct communication range. To achieve this, we define a generalized trust vector O_i that aggregates and propagates local trust information in a robust one-hop manner.

For vehicle i , the vector is defined as

$$O_i = [O_i(1), O_i(2), \dots, O_i(N)], \quad (33)$$

where O_i its current opinion toward all platoon members, and $O_i(j) \in [0, 1]$ represents the current trust opinion of vehicle i toward vehicle j .

1) Self-Trust. Each vehicle assumes full confidence in its own state:

$$O_i(i) = 1, \quad (34)$$

2) Direct-Neighbor Trust. For every neighbor $j \in \mathcal{N}_i$ with which vehicle i exchanges data directly, the trust opinion is set as :

$$O_i(j) = VT_{i,j}, \quad (35)$$

where $VT_{i,j}$ is the smoothed target-trust value derived in the previous subsection.

3) One-Hop Trust Propagation. For any non-neighbor vehicle $j \notin \mathcal{N}_i \cup \{i\}$, the host vehicle i estimates $O_i(j)$ using only information from its direct neighbors $\mathcal{N}_i$.

$$S_{i,j} = \left\{ O_k(j) \mid k \in \mathcal{N}_i, VT_{i,k} > \theta_{\min}, \delta_{i,k}(t) = 1 \right\} \quad (36)$$

Here, $\theta_{\min}$ is the minimum acceptable trust threshold,

Each valid neighbor k is associated with a *credibility weight*

$$w_{i,k} = VT_{i,k} \quad (37)$$

If the credible set $S_{i,j}$ is non-empty, the propagated opinion is computed using a *weighted median* operator:

$$O_i(j) = \text{wMed}(\{O_k(j)\}_{k \in S_{i,j}}, \{w_{i,k}\}_{k \in S_{i,j}}), \quad (38)$$

which limits the influence of outliers or manipulated values.

4) Index Distance-Based Fallback. If the credible set $S_{i,j}$ is empty, vehicle i resorts to a *host-centric distance weighting* of all its neighbors:

$$O_i(j) = \frac{\sum_{k \in \mathcal{N}_i} g_{i,k} O_k(j)}{\sum_{k \in \mathcal{N}_i} g_{i,k}}, \quad g_{i,k} = \frac{1}{1 + |i - k|}. \quad (39)$$

where $|i - k|$ is the index distance between vehicles i and k . The closer a neighbor k is to host i , the larger its influence $g_{i,k}$. This fallback guarantees that the trust vector remains fully populated even when no verified propagated opinion exists.

6) Final Expression. Combining all cases, the complete definition of the generalized trust vector is

$$O_i(j) = \begin{cases} 1, & j = i, \\ VT_{i,j}, & j \in \mathcal{N}_i, \\ \text{wMed}(\{O_k(j)\}, \{w_{i,k}\}), & S_{i,j} \neq \emptyset, \\ \frac{\sum_{k \in \mathcal{N}_i} g_{i,k} O_k(j)}{\sum_{k \in \mathcal{N}_i} g_{i,k}}, & S_{i,j} = \emptyset. \end{cases} \quad (40)$$

This generalized representation allows each vehicle to maintain a resilient, full-platoon trust profile that supports the adaptive observer weight design described in the following section.

5. WEIGHT DESIGN FOR OBSERVER BASED ON THE TRUST

The trust framework provides each vehicle with a set of quantified opinions $\mathcal{O}_{i,j}$ representing the reliability of information from other vehicles. This section shows how these values are used to define adaptive observer weights $w_{il}^j(t)$ and establishes conditions for stability of the estimation error dynamics.

5.1 Weight based on the trust

Based on the virtual communication graph $\bar{\mathcal{G}}^{(j)}$ as that includes the local observer node 0, each vehicle i identifies its legitimate neighbors at time t :

$$\mathcal{L}\mathcal{N}_i^{(j)}(t) = \begin{cases} \{l \in \mathcal{V} \mid \mathcal{O}_i(l) > 0.5\} \cup \{0\}, & \text{if } \mathcal{O}_i(j) > 0.5, \\ \{l \in \mathcal{V} \mid \mathcal{O}_i(l) > 0.5\}, & \text{if } \mathcal{O}_i(j) \leq 0.5, \end{cases} \quad (41)$$

To limit the influence of any single neighbor, a normalization factor is introduced.

$$n_{il}^{(j)}(t) = \max \left\{ \kappa, \left| \mathcal{L}\mathcal{N}_i^{(j)}(t) \right| + 1 \right\} \geq 1 \quad \text{for all } l \in \mathcal{L}\mathcal{N}_i^{(j)}. \quad (42)$$

where $\kappa > 0$ is the parameter to limit the maximum influence from the neighbors of i th vehicle. Then, the observer's weight gain can be chosen as

$$w_{il}^{(j)}(t) = \begin{cases} \frac{1}{n_{il}^{(j)}(t)}, & l \in \mathcal{L}\mathcal{N}_i^{(j)}(t) \\ 0, & \text{otherwise.} \end{cases} \quad (43)$$

$$w_{i0}^{(j)}(t) = \begin{cases} \frac{1}{n_{il}^{(j)}(t)}, & \mathcal{L}\mathcal{T}_j > 0.5 \\ 0, & \text{otherwise.} \end{cases} \quad (44)$$

According to the weight gain, the communication graph will be switched depending on the time. To ensure the connectivity of the communication graph, we list the following assumptions.

Assumption 3. (Jointly connected graph). A switching-directed graph is said to be jointly connected if there exists a positive constant $T > 0$ such that the union of the subgraphs over $[t, t + T]$, $\forall t \geq 0$ that is, $\cup_{[t, t+T]} \bar{\mathcal{G}}^{(j)}(t)$, contains a spanning tree.

5.2 Stability of the error

In this subsection, we explore the conditions that guarantee the stability of the estimation error dynamics of the proposed observer, as given in (8).

$$e_0^{(j)}(t+1) = (A_b - F_j C_b) e_0^{(j)}(t) - d_0^j, \quad (45)$$

6. SIMULATION

6.1 Objectives of the Simulation

The simulation study evaluates the proposed trust-based distributed observer under various cyberattack scenarios in a connected vehicle platoon. The main goals are to:

- Assess resilience of the observer against falsified or delayed information.
- Analyze how trust scores evolve to isolate malicious nodes.
- Compare estimation performance with and without local trust integration.
- Identify which types of attacks most affect estimation accuracy.

6.2 Simulation Setup

The simulation environment consists of a connected vehicle platoon composed of four vehicles. Each vehicle uses a third-order discrete-time longitudinal dynamics model.

Key simulation parameters such as control gains, trust decay factors, and attack signal patterns are presented in Tables I to V. The simulation runs are performed in MATLAB data.

The parameters of the CACC system and its controller are given in Table 1.

Parameter	Value	Description
(k_s, k_v, k_a)	2.0, 2.0, 1.0	CACC control gains
$(\alpha, \beta, v_0, \delta, T, s_0)$	2, 3, 33, 4, 0.4, 8	IDM parameters
(w_v, w_d, w_a)	1.2, 2.0, 0.9	Local trust weights
(β, C, k)	0.5, 0.2, 5	Trust decay & regularization
τ	1.5 s	Engine lag constant

Table 1. Key simulation parameters for the CACC system and trust framework.

6.3 Attack Scenarios

To assess the robustness of the vehicle platooning system, we define four categories of cyber-attacks: Bogus Messages, Replay/Delay Attacks, Collusion Attacks, and Denial-of-Service (DoS) Attacks. Each category includes specific cases implemented in the simulation environment to evaluate their impact on platoon stability and safety. These scenarios are detailed below. We suppose we know the controller of other to show the effectiveness of our observer.

Attack Type	Position Offset (m)	Velocity Offset (m/s)	Acceleration Offset (m/s ²)
Constant	± 5	± 2.5	± 0.2
Linear	$\pm 0.5t$	$\pm 0.2t$	$\pm 0.05t$
Sinusoidal	$\pm 5 \sin(0.5t)$	$\pm 2.5 \sin(0.5t)$	$\pm 0.2 \sin(0.5t)$

Table 2. Attack scenarios and their parameter offsets.

7. LIMITATION - CONCLUSION

The proposed trust-based observer effectively protects the platoon by isolating malicious agents and preventing the propagation of corrupted data. However, it does not

$$e_{\mathcal{DO}}^{(j)}(t+1) = \left(w_0^{(j)} \otimes A_b \right) e_0^{(j)}(t) + \left((I_N - \bar{\mathcal{L}}^{(j)}) \otimes A_b \right) e_{\mathcal{DO}}^{(j)}(t) - d_{\mathcal{DO}}^{(j)},$$

where

$$\bar{\mathcal{L}}^{(j)} = \begin{bmatrix} \sum_{l \in \{\mathcal{N}_1 \cup \{0\}\}} w_{1l}^{(j)} & -w_{12}^{(j)} & \cdots & -w_{1N}^{(j)} \\ -w_{21}^{(j)} & \sum_{l \in \mathcal{N}_2} w_{2l}^{(j)} & \cdots & -w_{2N}^{(j)} \\ \vdots & \vdots & \ddots & \vdots \\ -w_{N1}^{(j)} & -w_{N2}^{(j)} & \cdots & \sum_{l \in \mathcal{N}_N} w_{Nl}^{(j)} \end{bmatrix}$$

where $e_{\mathcal{DO}}^{(j)}(t) = \text{col} \left(e_1^{(j)}(t), \dots, e_N^{(j)}(t) \right)$.

$w_0^{(j)} = \text{col} \left(w_{10}^{(j)}, \dots, w_{N0}^{(j)} \right)$ is a vector to describe the connection between the local observer $\mathcal{LO}_j$ and other distributed observers $\mathcal{DO}_i$, $i \in \mathcal{V}$. $\bar{\mathcal{L}}^{(j)}$ is the virtual weighted Laplacian matrix of the graph $\bar{\mathcal{G}}^{(j)}$.

Then, the following theorem illustrates the primary sufficient condition to stabilize the error system (45).

Theorem 1. Consider a vehicle platoon (5) over a directed weighted communication network $\bar{\mathcal{G}}^{(j)}$. Under the observer proposed in (8), with the following conditions can be satisfied simultaneously:

- (1) There exists F_j such that $\rho(A_b - F_j C_b) < 1$;
- (2) For $i \in \mathcal{V}$, $w_{i0}^{(j)} \geq 0$ and $w_{il}^{(j)} \geq 0$ such that $\sum_{l \in \bar{\mathcal{G}}^{(j)}} w_{il}^{(j)} \leq 1$.
- (3) There exists $\alpha \in (0, 1)$ and a matrix norm $\|\cdot\|_*$ such that, uniformly over all admissible switches,

$$\| (I_N - \bar{\mathcal{L}}^{(j)}) \otimes A_b \|_* \leq \alpha < 1. \quad (47)$$

such that, uniformly over all admissible switches,

The estimated error (45) is input-to-state stable. In particular, there exist constants $c_0, c_1 > 0$ and $\lambda \in (0, 1)$ (independent of t) such that,

$$\|e_0^{(j)}(t)\| \leq \lambda^t \|e_0^{(j)}(0)\| + c_0 \bar{d}, \quad (48a)$$

$$\|e_{\mathcal{DO}}^{(j)}(t)\| \leq \lambda^t \|e_{\mathcal{DO}}^{(j)}(0)\| + c_1 (\|e_0^{(j)}(0)\| + \bar{d}), \quad (48b)$$

Proof. [sketch] **Local loop.** Choose $P = P^\top > 0$ and $Q = Q^\top > 0$ with $A_\ell^\top P A_\ell - P = -Q$. Then

$$V_0(t+1) - V_0(t) \leq -\lambda_{\min}(Q) |e_0^{(j)}(t)|^2 + 2|P A_\ell| |e_0^{(j)}(t)| |d^{(j)}(t)| + |P| |d^{(j)}(t)|^2,$$

which is a standard discrete-time ISS inequality implying the first bound and the ultimate bound on $e_0^{(j)}$.

Distributed loop. Let $W := I_N \otimes P$. Using (47b'),

$$V_{\mathcal{DO}}(t+1) = e_{\mathcal{DO}}^\top \left((I - \bar{\mathcal{L}}^{(j)}) \otimes A_b \right)^\top W \left((I - \bar{\mathcal{L}}^{(j)}) \otimes A_b \right) e_{\mathcal{DO}} + 2e_{\mathcal{DO}}^\top \left((I - \bar{\mathcal{L}}^{(j)}) \otimes A_b \right)^\top W (w_0^{(j)} \otimes A_b) e_0 + \text{terms in } |e_0^{(j)}|^2 \text{ and } |d^{(j)}|^2.$$

By (47), $\|(I - \bar{\mathcal{L}}^{(j)}) \otimes A_b\| \leq \alpha < 1$ for all t , so the homogeneous part contracts. The cross term is bounded via Cauchy-Schwarz and yields an affine ISS bound in $|e_0^{(j)}|$ and $|d^{(j)}|$. Since $e_0^{(j)}$ is already ISS w.r.t. $d^{(j)}$, the cascade is ISS, which gives the stated inequalities and ultimate bounds.

allow accurate estimation of the states of those malicious vehicles themselves. Once a vehicle's trust score falls below the threshold, its information is excluded from neighbor updates and is no longer corrected by trusted nodes. Consequently, its state estimate cannot converge to the true state. The framework thus prioritizes platoon safety and resilience over reconstructing the exact behavior of compromised vehicles a limitation shared by most existing distributed observer designs.

Conclusion and Future Work

This paper presented a distributed observer-based platoon control framework capable of maintaining stability and safety under cyber-attacks. The approach combines local and distributed observers with a trust evaluation mechanism to assess the reliability of inter-vehicle data, ensuring robust operation even with compromised nodes.

Future work will focus on enhancing the trust mechanism by coupling it more closely with the controller, enabling trust evaluation for non-neighboring vehicles, and improving adaptability under dynamic communication topologies. We will also investigate machine-learning-based trust estimation and extend the observer design by using matrix gains to improve estimation accuracy and convergence.